

A.1 Kinematics

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **26 marks**

Take $g = 9.81 \text{ m s}^{-2}$ and ignore air resistance unless told otherwise. Show your working.

Question 1 — The airport lap

(6 marks)

Waiting for a delayed flight, a traveller walks a complete lap of a rectangular terminal, 120 m by 80 m, in 400 s, ending exactly where she began.

- (a) Distinguish between distance and displacement, using this lap as your example, and state the value of each. [3]
- (b) Hence write down her average speed and her average velocity for the lap. [2]
- (c) Explain why her *instantaneous* speed at some moment can differ from the average you calculated. [1]

Question 2 — The pizza drone

(11 marks)

A delivery drone flies horizontally at a steady 12 m s^{-1} , at a height of 45 m, and releases a (well-packaged) pizza.

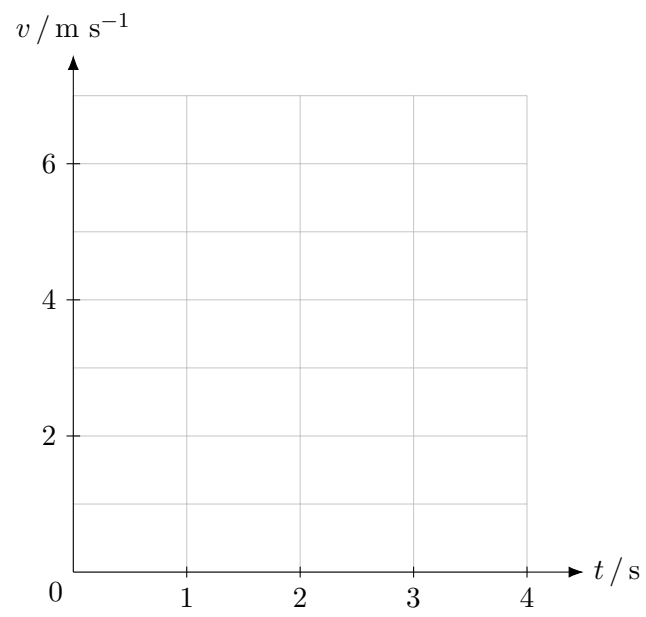
- (a) State what happens to the pizza's horizontal velocity after release, and why. [2]
- (b) Calculate the time the pizza takes to reach the ground. [3]
- (c) Calculate the horizontal distance it travels before landing. [2]
- (d) Show that it lands with a speed of about 32 m s^{-1} . [3]
- (e) A bystander claims the pizza “fell straight down from the moment it was dropped”. State the shape of the actual path seen from the ground. [1]

Question 3 — The rolling skateboard

(9 marks)

A skateboard is released from rest on a long, uniform ramp and accelerates uniformly. After 2.0 s it has travelled 3.0 m.

- (a) Calculate its acceleration. [2]
- (b) A classmate predicts it will have travelled 6.0 m after 4.0 s, “twice the time, twice the distance”. Calculate the actual distance and identify the error in the prediction. [3]
- (c) Sketch the velocity–time graph for the first 4.0 s, and state what feature of the graph gives (i) the acceleration and (ii) the distance travelled. [4]



Answers

Question 1 — The airport lap

- (a) Distance is the scalar length of the path travelled: $2(120 + 80) = 400$ m. Displacement is the vector from start to finish, which is *zero* because the lap closes.
- (b) Average speed $= 400/400 = 1.0 \text{ m s}^{-1}$; average velocity $= 0$ (zero displacement).
- (c) Her speed varies from moment to moment (queues, moving walkways); the average smooths those variations out.

Question 2 — The pizza drone

- (a) It stays at 12 m s^{-1} . No horizontal force acts (air resistance ignored), so there is no horizontal acceleration.
- (b) $45 = \frac{1}{2}(9.81)t^2 \Rightarrow t = \sqrt{90/9.81} = 3.0 \text{ s}$.
- (c) $x = 12 \times 3.03 = 36 \text{ m}$.
- (d) $v_y = 9.81 \times 3.03 = 29.7 \text{ m s}^{-1}$, so $v = \sqrt{12^2 + 29.7^2} = 32 \text{ m s}^{-1}$.
- (e) A parabola: the horizontal motion continues, so the path curves rather than dropping vertically.

Question 3 — The rolling skateboard

- (a) $s = \frac{1}{2}at^2 \Rightarrow 3.0 = \frac{1}{2}a(2.0)^2 \Rightarrow a = 1.5 \text{ m s}^{-2}$.
- (b) $s = \frac{1}{2}(1.5)(4.0)^2 = 12 \text{ m}$, not 6.0 m . Distance grows as t^2 : doubling the time *quadruples* the distance.
- (c) A straight line from the origin, $v = 1.5t$, reaching 6.0 m s^{-1} at 4.0 s . (i) The gradient gives the acceleration. (ii) The area under the line gives the distance travelled (here $\frac{1}{2} \times 4.0 \times 6.0 = 12 \text{ m}$, agreeing with (b)).